



Fig. 10: Display of app features on the screen

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

```
void handleCalculatorTouch(int x, int y) {
    int keyWidth = 35, keyHeight = 35;

    // Check which calculator button is touched
    int col = (x - 10) / (keyWidth + 5);
    int row = (y - 80) / (keyHeight + 5);

    if (col >= 0 && col < 4 && row >= 0 && row < 4) {
        int index = row * 4 + col;
        String calcKeys[16] = {"1", "2", "3", "+", "4", "5", "6", "-", "7", "8", "9", "*", "C", "0", "="};

        String keyPressed = calcKeys[index];

        if (keyPressed == "=") {
            // Calculate result
            text = calculateResult(text);
        } else if (keyPressed == "C") {
            text = "";
        } else {
            text += keyPressed;
        }

        // Update the result display area
        tft.fillRect(10, 30, 240, 40, TFT_GREEN);
        tft.setCursor(15, 50);
        tft.print(text + "_");
    }
}

void handleContactTouch(int x, int y) {
    // Add code to handle contact touch interactions
    // For example, selecting a contact for more details or making a call
}

String calculateResult(String expression) {
    // Here you can add a simple parser or use a library to evaluate the expression
    // For demonstration, we'll use a simple eval-like logic to handle basic operations
    float result = 0.0;
    char lastOp = '+';
    String currentNum = "";

    for (char c : expression) {
        if (isDigit(c)) {
            currentNum += c; // Build the current number
        } else {
            // Handle operator
            if (currentNum != "") {
                result = calculate(result, lastOp, currentNum);
                currentNum = "";
            }
            lastOp = c;
        }
    }
    // Final calculation
    if (currentNum != "") {
        result = calculate(result, lastOp, currentNum);
    }
    return String(result);
}
```

Fig. 11: App calculator screen UI code

the UI. The touch input handler activates UI actions in response to user input, while a separate function called 'calculate' performs the calculations and displays the results on the phone screen. Fig. 11 presents the app calculator screen UI code.

EFY note.

Further parts will be published in a coming issue, detailing upgrades to the phone with 4G LTE capabilities and additional functionalities. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/constructing-indusphone-part-2>

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